

Quick Setup Guide - Teachers & Teaching Events

Backend Quick Setup

1. Database Models ☒

- ☒ `Teacher.ts` model created with all required fields
- ☒ `TeachingEvent.ts` model created with all required fields
- ☒ Models exported from `src/models/index.ts`
- ☒ `UserRole.TEACHER` added to `User` model

2. Controllers ☒

- ☒ `teacher.controller.ts` - 8 public/protected endpoints
- ☒ `teaching-event.controller.ts` - 7 public/protected endpoints
- ☒ `admin.teacher.controller.ts` - 9 admin endpoints
- ☒ `admin.teaching-event.controller.ts` - 9 admin endpoints

3. Routes ☒

- ☒ `teacher.routes.ts` - public & protected routes
- ☒ `teaching-event.routes.ts` - public & protected routes
- ☒ `admin.teacher.routes.ts` - admin routes
- ☒ `admin.teaching-event.routes.ts` - admin routes
- ☒ Routes registered in `src/routes/index.ts`

4. Validation ☒

All routes include comprehensive validation:

- ☒ Express validator for input validation
- ☒ MongoDB ID validation
- ☒ Data type validation
- ☒ Required field validation

Frontend Quick Setup

1. Teacher Components ☒

- ☒ `TeacherCard.tsx` - Display teacher in grid/list view
- ☒ `TeacherProfile.tsx` - Full teacher profile page
- ☒ `TeachersList.tsx` - Teachers list with filters

2. Teaching Event Components ☒

- ☒ `TeachingEventCard.tsx` - Display event in grid/list
- ☒ `TeachingEventsList.tsx` - Events list with filters
- ☒ `TeachingEventDetail.tsx` - Detailed event view

3. TypeScript Types ☒

- ☒ `src/types/teacher.ts` - All interfaces for Teacher and TeachingEvent

4. Component Exports ☒

- ☒ `src/components/teacher/index.ts` - Teacher component exports
- ☒ `src/components/teaching-event/index.ts` - Teaching event component exports

API Endpoints Overview

Teachers

GET	/api/teachers	- List teachers (public)
GET	/api/teachers/:id	- Get teacher (public)
POST	/api/teachers	- Create profile (protected)
PUT	/api/teachers/:id	- Update profile (protected)
POST	/api/teachers/:id/reviews	- Add review (protected)
GET	/api/teachers/:id/events	- Get teacher's events (protected)
DELETE	/api/teachers/:id	- Delete profile (protected)

Teaching Events

GET	/api/teaching-events	- List events (public)
GET	/api/teaching-events/:id	- Get event (public)
POST	/api/teaching-events	- Create event (teachers only)
PUT	/api/teaching-events/:id	- Update event (teachers only)
POST	/api/teaching-events/:id/enroll	- Enroll (protected)
POST	/api/teaching-events/:id/unenroll	- Unenroll (protected)
POST	/api/teaching-events/:id/reviews	- Add review (protected)
DELETE	/api/teaching-events/:id	- Delete event (teachers only)

Admin Teachers

GET	/api/admin/teachers	- List all (admin)
POST	/api/admin/teachers	- Create (admin)
GET	/api/admin/teachers/:id	- Get details (admin)
PUT	/api/admin/teachers/:id	- Update (admin)
PUT	/api/admin/teachers/:id/approve	- Approve (admin)
PUT	/api/admin/teachers/:id/reject	- Reject (admin)
PUT	/api/admin/teachers/:id/featured	- Toggle featured (admin)
DELETE	/api/admin/teachers/:id	- Delete (admin)
GET	/api/admin/teachers/stats	- Statistics (admin)
PUT	/api/admin/teachers/bulk	- Bulk update (admin)

Admin Teaching Events

GET	/api/admin/teaching-events	- List all (admin)
POST	/api/admin/teaching-events	- Create (admin)
GET	/api/admin/teaching-events/:id	- Get details (admin)
PUT	/api/admin/teaching-events/:id	- Update (admin)
PUT	/api/admin/teaching-events/:id/approve	- Approve (admin)
PUT	/api/admin/teaching-events/:id/reject	- Reject (admin)
PUT	/api/admin/teaching-events/:id/featured	- Toggle (admin)
DELETE	/api/admin/teaching-events/:id	- Delete (admin)
GET	/api/admin/teaching-events/stats	- Statistics (admin)
PUT	/api/admin/teaching-events/bulk	- Bulk update (admin)

Integration Steps

1. Backend Integration

Install dependencies (if not already installed):

```
cd backend
npm install express-validator
```

Models are ready - No additional setup needed

Routes are registered - The new routes are automatically included in the main routes index

2. Frontend Integration

Create pages for Teachers:

```
// pages/TeachersPage.tsx
import { TeachersList } from '@components/teacher';

export default function TeachersPage() {
  return <TeachersList onSelectTeacher={(id) => navigate(`/teachers/${id}`)} />;
}
```

Create pages for Teaching Events:

```
// pages/TeachingEventsPage.tsx
import { TeachingEventsList } from '@components/teaching-event';

export default function TeachingEventsPage() {
  return <TeachingEventsList onSelectEvent={(id) => navigate(`/teaching-events/${id}`)} />;
}
```

Add routes to your router:

```
import TeachersPage from '@pages/TeachersPage';
import TeachingEventsPage from '@pages/TeachingEventsPage';
import TeacherDetailsPage from '@pages/TeacherDetailsPage';
import TeachingEventDetailsPage from '@pages/TeachingEventDetailsPage';

const routes = [
  // ... existing routes
  { path: '/teachers', element: <TeachersPage /> },
  { path: '/teachers/:id', element: <TeacherDetailsPage /> },
  { path: '/teaching-events', element: <TeachingEventsPage /> },
  { path: '/teaching-events/:id', element: <TeachingEventDetailsPage /> },
];
```

Features Implemented

Teacher Module

- ☒ Teacher profile creation with qualifications
- ☒ Verification and approval workflow
- ☒ Rating and review system
- ☒ Search and filtering (by subject, location, price, rating)
- ☒ Featured teacher management
- ☒ View count tracking
- ☒ Teaching methods specification (online/offline/hybrid)
- ☒ Certifications management
- ☒ Languages spoken
- ☒ Soft delete functionality

Teaching Event Module

- ☒ Event creation with detailed information
- ☒ Schedule management with multiple class times
- ☒ Student enrollment with capacity management
- ☒ Enrollment status tracking
- ☒ Rating and review system
- ☒ Search and filtering (by subject, method, difficulty, location, price)
- ☒ Learning objectives and prerequisites
- ☒ Materials list
- ☒ Certificate offering
- ☒ FAQ management
- ☒ SEO meta tags
- ☒ Meeting link for online classes
- ☒ Soft delete functionality

Admin Features

- ☒ Teacher approval/rejection workflow
- ☒ Event approval/rejection workflow
- ☒ Featured management for both
- ☒ Statistics and analytics
- ☒ Bulk operations
- ☒ Comprehensive filtering and search
- ☒ Soft and permanent deletion options

Frontend Features

- ☒ Responsive grid layouts
- ☒ Advanced filtering
- ☒ Pagination

- ☒ Loading states
- ☒ Empty states
- ☒ Modal dialogs
- ☒ Tab navigation
- ☒ Star ratings
- ☒ Real-time search
- ☒ Enrollment management
- ☒ Review submission

Testing Recommendations

Backend Testing

1. Test teacher creation with various roles
2. Test teaching event creation and filtering
3. Test enrollment and capacity management
4. Test approval workflows
5. Test rating calculations
6. Test soft and hard deletion
7. Test bulk operations

Frontend Testing

1. Test component rendering
2. Test search and filtering functionality
3. Test pagination
4. Test enrollment flows
5. Test responsive design on mobile
6. Test error handling
7. Test loading states

Documentation Files

-  `TEACHER_TEACHING_EVENTS_GUIDE.md` - Comprehensive guide (this file)
-  API documentation available in Postman collection
-  TypeScript interfaces well-documented in code

Next Steps

1. **Create admin pages** for teacher and teaching event management
2. **Implement payment system** for class bookings
3. **Add live video integration** for online classes
4. **Create certification system** for course completion
5. **Build student progress tracking**
6. **Implement scheduling calendar**
7. **Add notification system** for enrollment/class updates
8. **Create attendance tracking**
9. **Build analytics dashboard** for teachers

Common Issues & Solutions

Issue: Routes not working

Solution: Ensure routes are properly imported and registered in `src/routes/index.ts`

Issue: Models not found

Solution: Check that models are exported in `src/models/index.ts`

Issue: Validation errors

Solution: Review express-validator middleware setup in routes

Issue: Frontend components not rendering

Solution:

- Check import paths

- Verify TypeScript types
- Check for missing dependencies (lucide-react, react-router-dom)

File Checklist

Backend ☒

- [X] src/models/Teacher.ts
- [X] src/models/TeachingEvent.ts
- [X] src/controllers/teacher.controller.ts
- [X] src/controllers/teaching-event.controller.ts
- [X] src/controllers/admin.teacher.controller.ts
- [X] src/controllers/admin.teaching-event.controller.ts
- [X] src/routes/teacher.routes.ts
- [X] src/routes/teaching-event.routes.ts
- [X] src/routes/admin.teacher.routes.ts
- [X] src/routes/admin.teaching-event.routes.ts
- [X] Updated src/routes/index.ts
- [X] Updated src/models/index.ts
- [X] Updated User model with TEACHER role

Frontend ☒

- [X] src/components/teacher/TeacherCard.tsx
- [X] src/components/teacher/TeacherProfile.tsx
- [X] src/components/teacher/TeachersList.tsx
- [X] src/components/teacher/index.ts
- [X] src/components/teaching-event/TeachingEventCard.tsx
- [X] src/components/teaching-event/TeachingEventsList.tsx
- [X] src/components/teaching-event/TeachingEventDetail.tsx
- [X] src/components/teaching-event/index.ts
- [X] src/types/teacher.ts

Documentation ☒

- [X] TEACHER_TEACHING_EVENTS_GUIDE.md
- [X] TEACHER_TEACHING_EVENTS_QUICK_SETUP.md (this file)

Support

For questions or issues, refer to:

1. Main documentation: `TEACHER_TEACHING_EVENTS_GUIDE.md`
2. Similar implementations: Events, Vendors, Employees modules
3. Backend README: `backend/README.md`
4. Frontend README: `frontend/README.md`